

Assignment 1: Print Pattern through multiple logics



Logic 1:

Assume size to be the width at middle, and print the pattern

Logic 2:

Assume a central point as in coordinate geometry, measure the equal distance of coordinate pairs from the point (radius). Print on the locus of the radius.

Logic 3:

Code for logic 1:
StarPattern1.java

```
import java.util.Scanner;

class StarPattern1 {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int size, line, mid;
        System.out.print("Enter the size of pattern (width at the middle) :");
        size = scanner.nextInt();
        mid = size/2;
        for(line = 0; line<size;line++){
            if(line==0||line==(size-1)){
                System.out.println(" ".repeat(mid)+"*");
            }
            else{
                System.out.println(" ".repeat(Math.abs(mid-line))+"*"+
                ".repeat(size - 2*Math.abs(mid-line) -2)+"*");
            }
        }
    }
}
```

Output:

```
C:\Users\shriw\OneDrive\Documents\JavaProgramming>javac StarPattern1.java

C:\Users\shriw\OneDrive\Documents\JavaProgramming>java StarPattern1
Enter the size of pattern (width at the middle) :11

  *
 * *
*   *
*   * *
*   * *
*   * *
*   * *
*   * *
*   * *
*   * *
*   *
 * *
  *

C:\Users\shriw\OneDrive\Documents\JavaProgramming>
```

Logic 2: StarPattern2.java

```
import java.util.Scanner;

class StarPattern2 {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int size, centre;
        System.out.print("Enter the size of pattern (width at the middle) :");
        size = scanner.nextInt();
        centre = size/2;
        for(int i = 0; i < size; i++){
            for(int j = 0; j < size; j++){
                if(Math.abs(centre-i)+Math.abs(centre-j)==centre){
                    System.out.print("*");
                }
                else{
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

Output:

```
C:\Users\shriw\OneDrive\Documents\JavaProgramming>javac StarPattern2.java
C:\Users\shriw\OneDrive\Documents\JavaProgramming>java StarPattern2
Enter the size of pattern (width at the middle) :11
  *
 * *
*   *
*   *
*   *
*   *
*   *
*   *
*   *
*   *
 * *
  *

C:\Users\shriw\OneDrive\Documents\JavaProgramming>
```

Logic 3:
StarPattern3.java

```
import java.util.Scanner;
public class StarPattern3{
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size: ");
        int size = scanner.nextInt();
        if (size % 2 == 0) size++;
        int midPoint = size / 2;
        for (int currentRow = 0; currentRow < size; currentRow++) {
            int offset = (currentRow <= midPoint)
                ? currentRow
                : (size - 1 - currentRow);
            int leftBoundary = midPoint - offset;
            int rightBoundary = midPoint + offset;

            for (int currentCol = 0; currentCol <= rightBoundary;
currentCol++) {
                if (currentCol == leftBoundary || currentCol ==
rightBoundary) {
                    System.out.print("*");
                } else {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
        scanner.close();
    }
}
```

Output:

```
C:\Users\shriw\OneDrive\Documents\JavaProgramming>javac StarPattern3.java
```

```
C:\Users\shriw\OneDrive\Documents\JavaProgramming>java StarPattern3
```

```
Enter size: 11
```

A 16-pointed star shape formed by asterisks. The shape is symmetrical and has 16 points. The points are arranged in a circular pattern, with 16 asterisks forming the outer points and 16 asterisks forming the inner points. The points are arranged in a circular pattern, with 16 asterisks forming the outer points and 16 asterisks forming the inner points.

```
C:\Users\shriw\OneDrive\Documents\JavaProgramming>
```

Page No. ()

Date :

*****Practical 87*****

Java - ISE - 1 - 2024BIT038

a) a) jshell> int a = 'A'; a + 4

69

since 'A' is automatically stored as 65 in
int a,

→ a = 65

a + 4 = 69

b) - Assume a = 5

jshell> -a -++a+ --a

-6

once a is declared to be 5, it can't be made -5 by
just -a,

⇒ -5 - [++5] + [--6]

= -5 - 6 + 5

= -6

c) String b = "Alt b_c \sd";

jshell> b.replace(" ", "");

"Alt bcd"

It is for 'tab' formatting, used for leaving tab or 8
spaces, and '\s' for single space. However, 1 tab is
considered different than simple single space " ",
So, replace(" ", "") won't work on it

*****Printers
Nanded

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1) Byte.SIZE + Short.SIZE / Short.BYTES

16

SIZE method gives number of bits, BYTES gives number of bytes.

As compiler reads from left to right but precedence of operators ~~in from~~ decide how compiler works

$$\rightarrow 8 + 16/2$$

$$= 8 + 8 = 16$$

2) b - ?

 $\rightarrow$ (short) 5022026Range of short $\Rightarrow$ -32,768 to 32,767

$$5022026 - 32767 = 49,89,259$$

$$32768 \times 150 = 49,15,200$$

$$4989259 - 4915200 = 74,059$$

$$32768 \times 2 = 65536$$

$$74059 - 65536 = 8523$$

$$8523 - 32,768 = -24,245$$

$$-24245 - 1 \text{ (-ve number)}$$

$$\Rightarrow \text{(short)} 5022026 = -24246$$

***** Printers Nanded

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2.c)-?

→ No, `isBlank()` and `isEmpty()` methods can not be used replacably in `String` class, as they both work on different parameters of a string.

Example cases:

```
jshell> String s=""; a=" "; l="Hello";
```

```
s=>"";
```

```
a=>" ";
```

```
l=>"Hello";
```

```
s.isEmpty();
```

```
true
```

```
s.isBlank();
```

```
true
```

Since, there is no element in `s`, it is both Blank & Empty

```
a.isEmpty();
```

```
false
```

```
a.isBlank();
```

```
true
```

Since, there is single ~~to~~ whitespace element in `s`, it is not Empty, but it does not have anything except Blank space, `isBlank()` returns true.

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`l.isEmpty();``false``l.isBlank();``false`

Since, `l` is 'Hello', it has non-zero length, so `isEmpty()` returns false. Also, it does not have only blank spaces, `isBlank()` returns false too.

$\Rightarrow$ `isEmpty()` is used there to determine if length of string is zero.

`isBlank()` is used to determine whether there are elements except the white space.

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227-2

→ To print pattern:

```

X   X       X   X
X   X       X
X   X       X   X
X   X       size 3
X   X

```

The entered value 4 is not
odd. Please enter an odd
size
→ For even size.

size 5

```

void printPattern (int size) {
    if (size % 2 == 0) {
        System.out.println("The entered value " + size + " is not  
odd. Please enter an odd size.");
    } else {
        int mid = size / 2;
        for (int line = 0; line < size; line++) {
            if (line < mid) {
                System.out.println(" ".repeat(line) + "X" + " ".  
repeat(size - 2 * (line + 1)) + "X");
            } else if (line == mid) {
                System.out.println(" ".repeat(line) + "X");
            } else {
                System.out.println(" ".repeat(line - mid) + "X" +  
" ".repeat(2 * (line - mid) + 1) + "X");
            }
        }
    }
}

```

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